CODE HEALTH CHECK · 5 NOVEMBER 2026 · SOFC POC

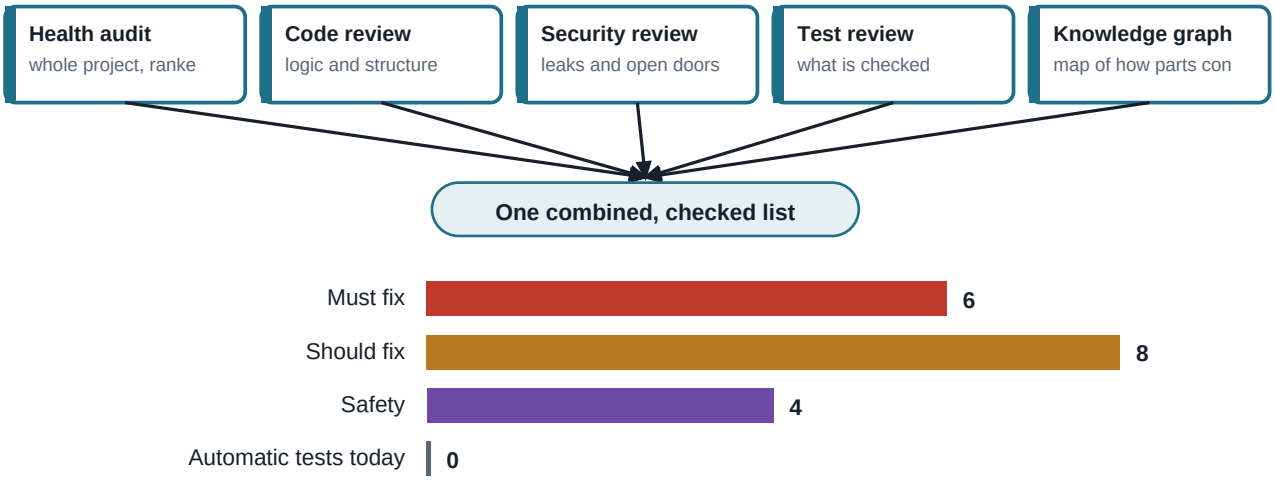
Is the SOFC analysis code healthy?

Contents: 1 What was checked · 2 Where numbers can go wrong · 3 Safety and tests · 4 Project map · 5 What to do next

Mostly yes. The stage rules are carefully written and every threshold says where it came from. But **six places can give a wrong number or stop with an error**, nothing is checked automatically, and two small local apps can be opened from the office network.

Why it matters: the stage labels feed the availability figures, the slopes, the twin and the synthetic data. A small slip early in that chain shows up in every report after it.

What ran, and what it found



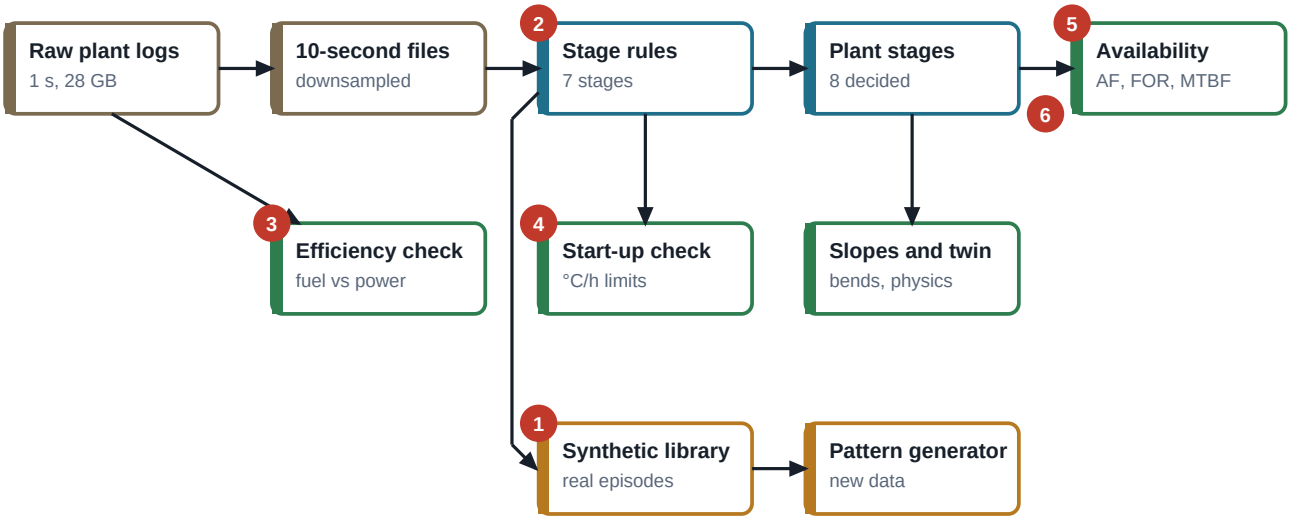
Five checks read the 31 code files, 5 documents and 3 drawings (the 28 GB of raw plant data was not read). Every must-fix item was traced in the code by me or by the code reviewer.

IF YOU REMEMBER ONE THING

The rules are sound, but six spots can quietly bend the numbers. Fix the gas signal question and the trip-zone bug first.

Where numbers can go wrong

Six spots can give a wrong number or stop with an error. Each red dot sits where the problem enters the chain.



Read left to right: raw logs become minutes with a stage, then reports. A problem early in the chain (dot 2, the gas signal) reaches everything to its right.

Must fix

1 **Trip moments can leak into the synthetic training library** [traced by hand]

`Synthetic data generation/build_library.py, lines 158–163`

When two safety events are less than 20 minutes apart (a purge followed by a stop), the code picks a 'clean' stretch that sits inside the trip. The generator then learns trip behaviour as if it were normal.

Fix: Join the ±10-minute danger zones first, then take the longest gap between them. Rebuild the library.

2 **The 'natural gas' signal is really the POx air flow** [checked in code]

`data_analysis/scripts/operating_states.py, line 117`

The stage rules use FIRC 1,2 (POx air) as natural gas. The twin uses FIRC 1,4 (the real natural-gas flow). So the 'Fuel switch' stage means two different things in the analysis and in the twin.

Fix: Your decision: switch to FIRC 1,4, or keep FIRC 1,2 and call it a proxy everywhere.

3 **The efficiency check reads columns that go bad from 20 March** [checked in code]

`data_analysis/scripts/efficiency_check.py, line 11`

It uses the EIRC 4x voltage and current columns. operating_states.py warns these shift to the wrong place from 20 March about 11:00. Late-March efficiency numbers may be wrong.

Fix: Use the PLC monitor columns, the same ones the stage rules use.

4 **The start-up speed check can raise false alarms across logger gaps** [code reviewer]

`data_analysis/scripts/flow_check.py, lines 157–161`

It measures 'one hour' as 60 rows of data. If the logger stopped for 4 hours inside those rows, 4 hours of warming look like one hour and break the 120 °C/h limit.

Fix: Put the data on a regular 1-minute clock first, as stage_signals.py already does.

5 **The availability script crashes if the data ends during a recovery** [checked in code]

`data_analysis/scripts/availability.py, line 61`

After each trip it looks for the next Hot standby. If none exists yet, the script stops with an error. The 'lenient' rule also counts 16 forced minutes, not 15.

Fix: Skip a trip with no recovery yet; end the lenient window at 14 minutes after the cut.

6 **Trip time is counted three slightly different ways** [code reviewer]

`plant_stages.py 77–82 · industrial_stages.py 117 · availability.py 59`

Each file starts and ends the trip window one minute apart. Trip minutes, S8 minutes and forced hours then differ by 1–2 minutes per trip and never add up exactly.

Fix: One shared helper for the trip window, used by all three.

Should fix

These do not give wrong numbers today, but they make the next mistake likely.

7 **No automatic tests**

None of the rules above has a test that would fail when it breaks.

Fix: Write the three tests on page 3.

8 **Old saved results are reused without warning**

slopes_jerk.py, industrial_stages.py and pattern_filler.py read saved files. After a rule change they keep using old stage labels.

Fix: Rebuild a saved file when its source is newer.

9 **The synthetic generator still uses the old 7 stages**

The decided stages are the 8 plant stages. The library, splitter and generator are built on the old 7.

Fix: Your decision: move the generator to the 8 stages, or keep 7 on purpose.

10 **The bend and jerk method was copied three times and drifted**

slopes_jerk.py, twin_2d/tools/bends.py and app.js smooth differently, so the twin's bend panel differs from the analysis.

Fix: Let bends.py import the functions from slopes_jerk.py.

11 **Warming speed is blank at the start and end of each day file**

Rates are worked out one day at a time (operating_states.py 124–128), so the first and last 30 minutes have no value.

Fix: Work out the rates on the joined days.

12 **A logger drop-out can hide a hot trip**

trip_check.py 136 averages the stack temperature without removing the –1 drop-out value.

Fix: Ignore values at or below 0 before averaging.

13 **Same name, different meaning**

'Hot' is 450 °C in one file and 400 °C in another; an 'episode gap' is 2 min, 60 min or 6 h; two functions are both called trip_events.

Fix: Give each its own name, or keep shared values in one place.

14 **Near-copies of the same helper**

The 'episodes' helper is written 5 times and 'blocks' 3 times.

Fix: Merge the copies that follow the same rule.

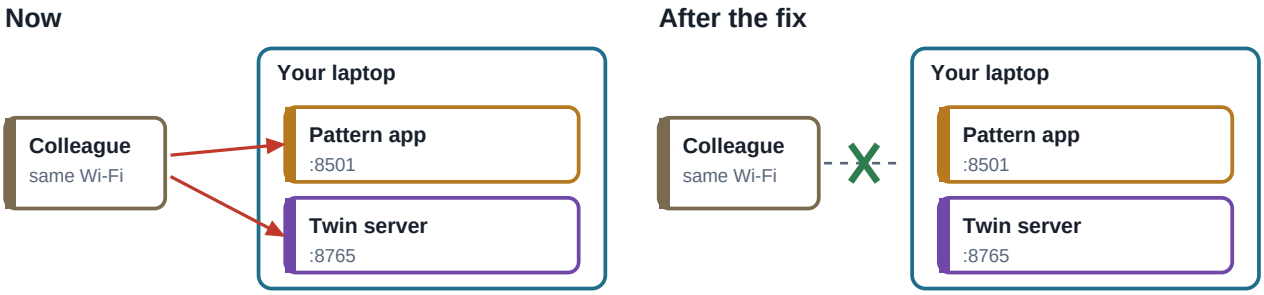
'traced by hand' = I followed an example through the code; 'code reviewer' = the review traced it and I did not re-run it.

IF YOU REMEMBER ONE THING

Dot 2 (which flow counts as natural gas) is a decision for you; the other five are plain code fixes.

Safety and tests

No serious security holes. The real risks are two local apps that open to the whole network, and real plant data in pages you might share.



Both apps listen on every network card by default. One setting each limits them to this computer (S1 and S2).

Safety items

S1 **The pattern app can be opened by anyone on the same network**

The Streamlit app listens on every network card. On office or hotel Wi-Fi a colleague can open it, see real episode tables and save files.

Fix: Add two lines to .streamlit/config.toml: [server] and address = "127.0.0.1".

S2 **The twin's start file shares its folder with the whole network**

start.bat runs a file server open to every network card, with folder listings.

Fix: Change the last line to: python -m http.server 8765 --bind 127.0.0.1

S3 **The published twin carries real plant data**

Sensor envelopes for 111 tags, PLC tag names, trip limits and a P&ID; redraw are in the page.

Fix: Keep the link private; ask the data owner before sharing outside the company.

S4 **Two PDFs show your Windows user name**

hs-cooling-slope-explained.pdf and twin-trip-rules-explained.pdf contain C:/Users/mrgur/... links.

Fix: Rebuild those two PDFs without the local links.

Checked and clean: No unsafe file loading (no pickle or eval); No shell commands built from text; No passwords, keys or IP addresses in code; The twin page makes no network calls; No outside scripts in the twin page.

Tests: none today, three to write first

A test is a small fake example with a known answer. If someone changes a rule by mistake, the test fails and says so.

Test	Fake example	Expected answer
Availability numbers	A 65-minute toy table: 10 min Part load, 5 min purge, 40 min cooling, 10 min Hot standby, one trip.	Strict rule gives 55 forced minutes; every stage name maps to a bucket.
Plant stage order	One row each: Shut-down with natural gas and air; Hot standby with natural gas and no current.	Air-flow cooling, not Fuel switch; then Fuel switch.

Trip windows	Safety cuts at 0, 25 and 50 minutes; then at 0 and 31 minutes.	One trip ending at 65 minutes; then two trips.

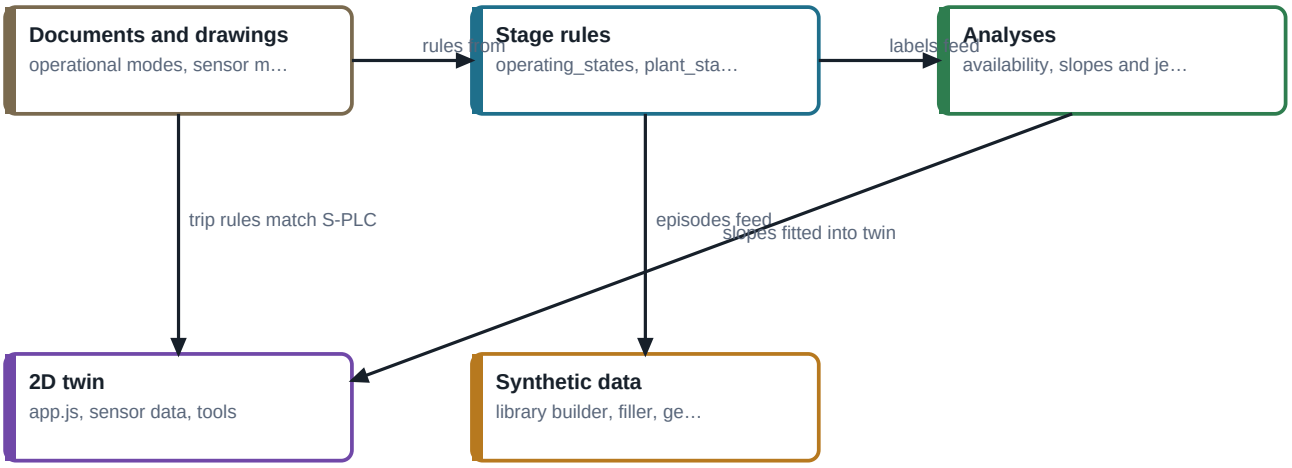
Not worth testing: plots, colours and the file loaders. For the twin, keep the 8-episode replay error (17.5 °C) as the number to watch.

IF YOU REMEMBER ONE THING

Two one-line changes close the network door. Three small tests would have caught items 1, 5 and 6.

Project map

The knowledge graph reads every code file, document and drawing and draws a line wherever one part uses or matches another. It found 634 parts, 1300 links and 27 natural groups.



The 27 groups fall into these five areas. The document 'Operational modes' is the strongest bridge between documents, stage rules and the start-up check.

Busiest parts

- segment() – builds one day's stages
- run_all() – runs every day
- fill() – fills a synthetic pattern
- generate() – makes a random pattern

Links found between documents and code

- The twin's trip purge matches the safety PLC (S-PLC) emergency purge
- The twin's Emergency stop button matches the S-PLC emergency stop
- The twin's cooling slope matches the measured ramp rates
- The twin's stage chain matches the start-up stage in the documents

Open the full graph: `01_POC/graphify-out/graph.html` (browser) and `GRAPH_REPORT.md`.

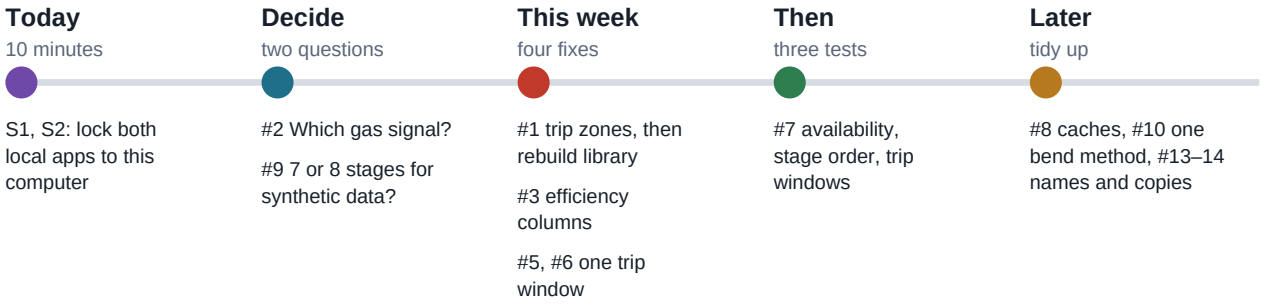
Left out on purpose: the 51 PDFs in deliverables, 69 plot images and the notebooks. The 6 Excel files were skipped (Excel reader not installed). 319 links point to libraries outside the project; the graph is still usable.

IF YOU REMEMBER ONE THING

Everything hangs off the stage rules. That is why a fix there (like the gas signal) matters more than a fix anywhere else.

What to do next

Start with the 10-minute network fix, answer two questions, then fix and protect the numbers.



Numbers refer to the items on page 2; S1 and S2 are on page 3.

Two decisions for you

2 **Which flow should count as natural gas?**

Switch the stage rules to FIRC 1,4 (the real natural-gas flow, as the twin does), or keep FIRC 1,2 (POx air) and label it as a stand-in everywhere.

Fix: My recommendation: switch to FIRC 1,4, then re-run plant_stages.py and compare the Fuel switch minutes before and after.

9 **Should the synthetic data use the 8 decided stages?**

The generator still builds on the old 7 stages, so its output does not use the names in the twin and the reports.

Fix: My recommendation: move it to the 8 stages when you next rebuild the library (after fix 1).

What I did not check

- The notebooks, except to see how the scripts are called
- Whether the six problems change today's published numbers (nothing was re-run on the plant data)
- Whether the clock change on 29 March 2026 shifts timestamps
- Installed package versions against known security issues

No project file was changed. New files: graphify-out/, .graphifyignore and this PDF.

IF YOU REMEMBER ONE THING

Lock the two apps today. Then answer the gas-signal question, because every stage and report depends on it.